Tugba Ates

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To build hands-on experience with KNIME, I independently designed and built the two end-to-end workflows summarized below — covering data cleaning, visualization, predictive modeling, live API integration, and text analysis. Both workflows are public on my KNIME Hub profile: hub.knime.com/tugbaates.

1 Spotify Tracks Analysis

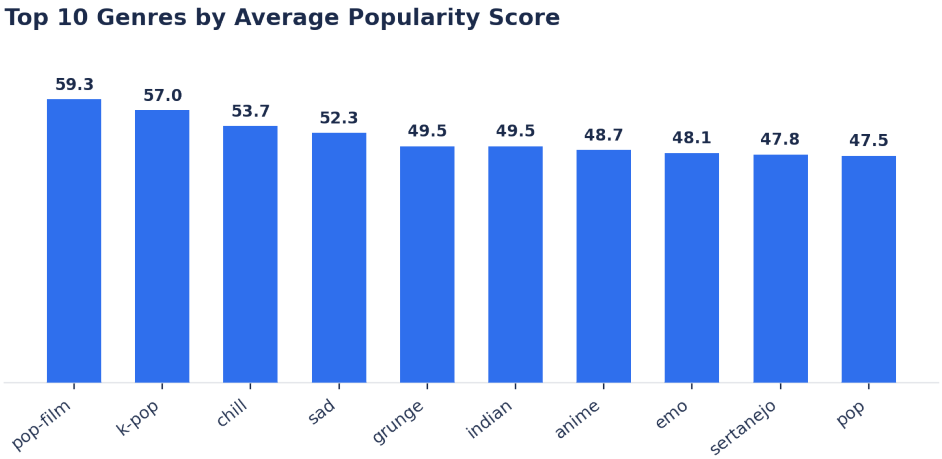
An end-to-end analysis of a 114,000-song Spotify dataset: cleaning, genre-level popularity comparison, audio-feature correlation analysis, and a Random Forest regression model that predicts a track's popularity from its audio features alone.

114,000	$R^2 = 0.54$	10.0
Tracks analyzed	Model accuracy	Avg. prediction error (pts)

PIPELINE



KEY FINDING



Pop-film, k-pop, and chill are the most consistently popular genres in the dataset. A correlation analysis of audio features also showed a strong positive relationship between *energy* and *loudness*, and a strong negative relationship between *energy* and *acousticness* — both musically intuitive and useful signals for the predictive model.

2 KNIME Community Forum Trend Analysis

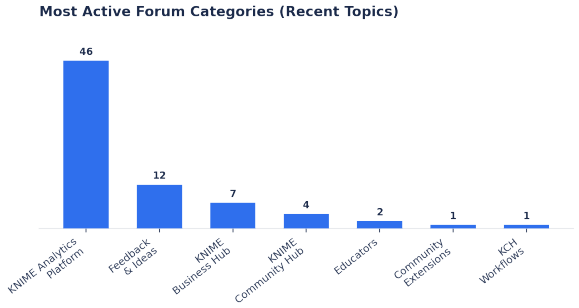
A live-data project built around KNIME's own public community forum: it fetches recent topics directly from the forum's API, parses the raw JSON into a clean table, and analyzes which categories are most active and which keywords trend in recent discussion titles — a small way of getting to know the community I'd like to join.

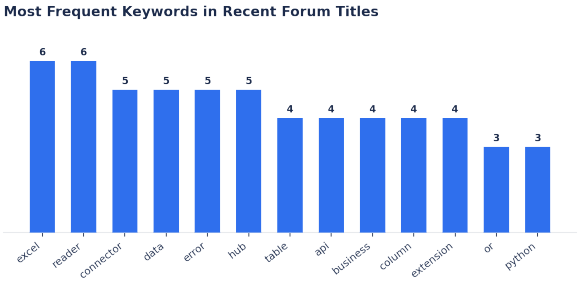
Live API	90	341
Real-time data source	Topics processed	Unique keywords extracted

PIPELINE



KEY FINDINGS





The **KNIME Analytics Platform** category dominates recent activity, and the most frequent keywords in topic titles — *excel, reader, connector, data, error, hub, python* — point to file I/O, connectors, and Python integration as the areas users engage with most.